

CSC 212: Data Structures and Abstractions

01: Introduction

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CSC 212

Learn to model and solve complex problems with computers.

The fundamentals that let you design, evaluate, and VERIFY algorithms

Course description

- Introduction to **fundamental data structures** and their algorithms
 - ✓ arrays, lists, stacks, queues, trees, hash tables, graphs (most popular topics for job interview questions)
 - ✓ survey of classic algorithms for sorting and searching
- Basic principles of analysis of algorithms
 - ✓ improve your foundation of CS theory
- Writing code that runs efficiently
 - ✓ choosing good algorithms and data structures

Presumes a strong foundation in C++ programming, including:

- Pointers
- Classes and objects
- Recursion

Why this course in the AI era?

- What does AI do well?
 - ✓ syntax, boilerplate, API recall, translating between languages
 - ✓ explaining a concept five different ways until one lands
 - ✓ decoding a 400-line C++ template error
- Where AI might be unreliable
 - ✓ choosing the right structure for a problem, given real constraints
 - ✓ asymptotic claims
 - ✓ C++ memory semantics: ownership, lifetimes, dangling pointers, copy costs
 - ✓ verification
- The job changed shape, not difficulty
 - ✓ adapting from coding => specifying, reviewing, and being accountable for it

Every system you will build, including AI systems, is made of data structures. Embeddings live in arrays. Vector search is a nearest-neighbor structure. A tokenizer is a trie. A KV-cache is a hash table with an eviction policy.

Course information

- Course Website
 - ✓ Syllabus, Schedule, Resources
 - ✓ <https://homepage.cs.uri.edu/~malvarez/teaching/csc-212/>

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What is expected from you?

- Attend lectures/labs
 - ✓ students are expected to attend all lectures and labs
 - ✓ regular attendance is linked to higher grades and better comprehension of course material
- Participate and think critically
 - ✓ ask questions (lectures, labs, office hours, Ed, ...)
- Start working on assignments early
 - ✓ refrain from simply copying and pasting answers generated by LLMs
- Laptops and cellphones are **NOT permitted** unless being used for taking notes

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Q1-1

- According to the syllabus, what happens to a homework assignment submitted one day late?
 - A. A 10% grade penalty is applied
 - B. It is not accepted
 - C. It is accepted with instructor approval only
 - D. It is accepted within a 24-hour grace period
 - E. None of the above

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Answer

- B) It is not accepted
 - ✓ late submissions are not accepted, instead, you get 6-9 days of lead time on every assignment, so start early and use office hours if you're stuck

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Q1-2

- To pass this class, which of these must be true, regardless of your final grade?
 - A. You attended at least 80% of lectures
 - B. Your assignment average is ≥ 50
 - C. Your average across all exams is ≥ 50
 - D. You scored ≥ 50 on the final exam alone
 - E. Not enough information

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Answer

- B) Your average across all exams is ≥ 50
 - ✓ exams are where the course verifies you actually learned the material

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Need a refresher on C++ programming?

- Pick a textbook (learn syntax)
- Solve Challenges



Kattis



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Q1-3

- What is wrong with this code?

```
void grow(int *arr, int n) {  
    arr = new int[2 * n];  
}  
  
int main() {  
    int *data = new int[10];  
    grow(data, 10);  
    data[15] = 42; // <-- what happens here?  
}
```

- A. Nothing, `grow` correctly reallocates the array
- B. `grow` receives a copy of `data`, so `data` still points to the old array, then `data[15]` writes out of bounds
- C. `new int[2*n]` is invalid C++, the size must be a compile-time constant
- D. The code leaks memory but is otherwise correct
- E. None of the above

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Answer

- B) `grow` receives a copy of `data`, so `data` still points to the old array, then `data[15]` writes out of bounds
 - ✓ assigning to `arr` rebinds the local copy and has no effect on `data` in main
 - ✓ consequences, in order of severity:
 - `data` still points to the original 10-element block, so `data[15]` writes past the end, undefined behavior, and it may not crash
 - the freshly allocated block is leaked, nothing points to it
 - neither block is ever `delete[]`-ed

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Q1-4

- What is the behavior of this program?

```
std::vector<int> v = {1, 2, 3, 4, 5};
int& r = v[0];
v.push_back(6);
std::cout << r;           // <-- what does this print?
```

- A. Prints 1, `r` refers to the first element, which is unchanged
- B. Prints 6
- C. Compile error, you cannot bind a reference to `v[0]`
- D. Undefined behavior, `push_back` may reallocate, leaving `r` dangling
- E. None of the above

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Answer

- D) Undefined behavior, `push_back` may reallocate, leaving `r` dangling
 - ✓ `push_back` may exceed the vector's capacity, in which case it allocates a new buffer, copies the elements over, and frees the old one
 - ✓ any reference, pointer, or iterator into the old buffer, including `r`, is left dangling
 - ✓ it might print 1. It might print garbage. It might work every time you test it and fail in production
- A) is so tempting: the value at index 0 really is unchanged. But `r` does not refer to 'index 0', it refers to a memory address, and that address is no longer yours

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Warming up

- Adjacent elements sum
 - ✓ find the **maximum sum** of any pair of adjacent elements in an array of integers

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11	3	5	3	2	-5	6	7	-9	-2	13	2
----	---	---	---	---	----	---	---	----	----	----	---

first pair

for this example, what is the maximum sum of two adjacent elements?

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Q1-5

- Complete the following program:

```
#include <vector>
#include <iostream>

int max_sum(std::vector<int>& nums) {
    // TODO
    return -1;
}

int main() {
    std::vector<int> nums = {1, -2, 3, 4, -1, 2, 1, -5, 4};
    int result = max_sum(nums);
    std::cout << "Maximum sum is: " << result << std::endl;

    return 0;
}
```

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Q1-6

- A student implements `max_sum` by initializing `best` with `-1`, then looping over adjacent pairs, keeping the largest sum. On which input does this return a wrong answer?

- A. It never fails, -1 is a safe starting value
- B. {-5, -3, -8, -3}, every pair sums to a negative number
- C. {0, 0, 0, 0}, every pair sums to zero
- D. {1, 2, 3, 4}, the pairs are positive
- E. None of the above

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Answer

- B) {-5, -3, -8, -3}, every pair sums to a negative number

- initializing `best = -1` assumes the answer is at least -1
- the bug is invisible on most inputs, it fires only when every adjacent pair sums below -1, so it survives casual testing indefinitely
- the fix**: always initialize from the data, not from a guess

```
int best = nums[0] + nums[1]; // must also handle n < 2 first
for (size_t i = 1 ; (i + 1) < nums.size() ; ++i)
    best = std::max(best, nums[i] + nums[i+1]);
```

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